

Life Expectancy Changes Due to Kidney Donation Across Race, Sex, and Age

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Abstract

Background. Living kidney donors face a small but well-documented elevation in long-term risk of end-stage renal disease (ESRD), yet this risk has not been translated into a decision-relevant life-expectancy metric stratified by the demographic factors known to modify it.

Methods. I constructed a Markov microsimulation model comparing lifetime outcomes for matched donor and non-donor cohorts, entering at the age of donation and simulated in annual cycles, parameterized from the United States Renal Data System (USRDS), the Scientific Registry of Transplant Recipients (SRTR), and published cohort studies of post-donation ESRD incidence and mortality.

Results. The life-expectancy implications of the published risk estimates depend primarily on which published estimate of the donor all-cause mortality hazard ratio (HR) is assumed: the three shorter-follow-up cohorts in the largest meta-analysis are consistent with HR near 1.0, while the one cohort followed long enough to detect a late-emerging effect reports HR = 1.30 with excess mortality apparent only after approximately a decade. Holding the HR fixed at any single published value places the cost between under two months (HR = 1.00) and more than four years (the upper confidence bound of the long-follow-up estimate). A time-varying base case encoding this decade-long onset delay yields a mean cost of 2.4 years. This base-case cost was 84% larger at age 25 than at age 55. Race and sex mattered less: Black donors' cost exceeded White donors' by 16% at age 25, but the difference disappeared by age 55, and male and female donors differed by about 2% at the base-case age of 40. Any other modeled parameter moves the result by a maximum of about five weeks. The life-expectancy benefit of prior-living-donor priority waitlist access was approximately 6 days unconditionally for the individual donor, and that of a National-Kidney-Registry-style voucher to designated non-donor family members was under two days on average; both were worth hundreds of days conditional on the designated beneficiaries actually developing ESRD.

Conclusions. These findings support age-, race-, and sex-specific life-expectancy disclosure during living-donor informed consent, indicate that priority-access mechanisms function as catastrophic insurance rather than as interventions expected to shift average outcomes, and highlight resolving the time-course of donor all-cause mortality risk as a research priority, since it dominates every other source of uncertainty in this analysis.

Keywords

living kidney donation, life expectancy, Markov model, end-stage renal disease, health disparities

Highlights

- The life-expectancy cost of kidney donation depends primarily on which published donor mortality hazard ratio (HR) is assumed: under two months at HR = 1.0 (no excess mortality) to over four years at the upper confidence bound of the longest-follow-up cohort. Under a time-varying base case that reconciles the existing cohort data (excess mortality emerging only after roughly a decade), the estimate is 2.4 years.
- This base-case cost is largest for the youngest donors. Black donors face a larger cost than White donors at young ages, but the difference disappears by age 55; men and women differ little at age 40, with the gap widening to 7–9% at ages 25 and 55.
- Resolving the time-course of post-donation mortality risk is the dominant research priority: whether and when excess mortality emerges determines the dominant share of the life-expectancy cost.

- Priority waitlist access benefits individual donors by approximately 6 days at the population level, and designated non-donor family members (via a National Kidney Registry voucher) by under two days on average; either benefit rises to roughly 164–1,488 days if kidney failure actually occurs, depending on age at onset.

Introduction

Living donor kidney transplantation (LDKT) remains the treatment of choice for end-stage renal disease (ESRD),

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offering better graft function and longer allograft survival than deceased-donor kidney transplantation (DDKT).¹ Yet the gap between organ need and organ supply persists: more than 4,000 candidates died on the US kidney waiting list in 2024, and a comparable number were removed for transplant-disqualifying deterioration, while the annual number of LDKTs has remained essentially flat for over a decade.² Expanding living donation therefore rests on two complementary levers: (1) recruiting more donors and (2) better understanding, quantifying, and managing the risk that donation poses to the donor.

The safety literature on living donation is large, but has historically emphasized relative rather than absolute, or life-expectancy-scaled, risk. Perioperative mortality is low (3.1 per 10,000 donors) but not uniform, rising sharply with male sex, Black race, and pre-existing hypertension.³ Longer-term follow-up using healthy, contraindication-screened non-donor comparators established that living donors face a several-fold higher risk of subsequently developing ESRD themselves, even though the absolute magnitude of that risk remains small.^{4,5} Whether this translates into excess all-cause mortality is unresolved, and the disagreement in the literature is not simply a matter of noisy estimates of one fixed effect.

To date, O’Keeffe et al. (2018)⁶ completed the largest meta-analysis of follow-up studies of long-term mortality among living kidney donors. They pooled four cohort estimates among donors relative to healthy matched controls: Segev et al. (2010),³ with a 6.3-year follow-up; Garg et al. (2012),⁷ with a 6.5-year follow-up; Berger et al. (2011),⁸ with a 10-year follow-up; and Mj  en et al. (2014),⁹ with a 15.1-year follow-up. Of those four studies, the only one to report a hazard ratio (HR) greater than 1.0 was that of Mj  en et al. (HR= 1.30, 95% CI 1.11–1.52). O’Keeffe et al. concluded, “We found no evidence that living kidney donors had higher risk for mortality.” Mj  en disagreed with this conclusion, arguing in a letter responding to O’Keeffe’s meta-analysis: “the survival curves in our study only started to separate at more than 10 years of follow-up, indicating that increased mortality after kidney donation may manifest only after long-term follow-up. At a follow-up of less than 10 years, the findings from these...studies are not contradictory. No published studies on kidney donors oppose our finding of increased mortality only after more than 10 years of follow-up.”¹⁰ O’Keeffe agreed.¹¹ No detectable excess mortality within the first post-donation decade, alongside a significant excess in the one cohort followed long enough to observe it, is consistent with a time-varying hazard rather than a single fixed effect that the literature estimates with varying precision, and is treated accordingly in this analysis.

Post-donation ESRD risk is not distributed evenly across the donor population. Black donors experience roughly three times the 15-year cumulative incidence of ESRD observed in White donors, a disparity that persists after matching to equally healthy non-donor comparators of the same race.⁴ Part of this excess risk has a defined biological correlate: two-copy *APOL1* renal-risk-variant genotypes, disproportionately carried by Black individuals, are among the strongest known risk factors for progression to chronic kidney disease (CKD) stage 3 or higher and have been directly implicated in the long-term kidney health of Black

living donors.¹² Donation itself is also not evenly distributed: women have consistently made up a majority of living donors, a pattern attributable in part to financial barriers to donation and disparities in access to unpaid leave that fall more heavily on lower-income men.¹³ Together, these findings motivate stratifying any life-expectancy analysis of donation by age, race, and sex rather than reporting a single population-average effect. **Donors who do develop ESRD also differ from other patients with kidney failure: they are listed earlier, are more often listed before starting dialysis, and have lower mortality after ESRD than matched non-donors,¹⁴ so outcomes measured in the general ESRD population cannot simply be transferred to them.**

Policy has responded to donor risk primarily by protecting donors’ future access to a transplant should they themselves develop ESRD. The Kidney Allocation System implemented in 2014 (KAS250) grants prior living donors priority access to the deceased-donor waiting list, and early evaluation confirmed that this priority meaningfully shortened prior donors’ time to transplant.¹⁵ Independently, a national registry analysis of transplant candidates who had previously donated an organ found that such priority access translated into substantially shorter waiting times and higher-quality kidneys compared with matched non-donor candidates, confirming that the policy achieves its intended effect at the level of the individual recipient.¹⁶ The National Kidney Registry (NKR) has extended this priority concept beyond the donor’s own future risk through its voucher program, under which a donor may designate family members, or in some cases any future intended recipient, to receive expedited, priority access to a living-donor kidney should they ever need one. Two large Monte Carlo simulation studies, spanning 50- and 100-year horizons, have shown that the voucher program is programmatically sustainable.^{2,17}

These two literatures, quantifying the donor’s own excess ESRD and mortality risk and demonstrating the programmatic success of priority-access mechanisms, have so far remained largely separate. The voucher simulations answer a supply-side question (will there be enough donors to honor future redemptions?) but do not quantify the demand-side question of how much life expectancy a voucher, or PLD priority access more generally, is actually worth to the person who holds it. Similarly, the risk literature reports relative risks and cumulative incidences but stops short of translating these into an integrated, individual-level life-expectancy trade-off that simultaneously accounts for the donor’s elevated risk and demographic heterogeneity. **This paper addresses that gap by asking what the published risk estimates imply for individual donors: using a Markov microsimulation model parameterized entirely from existing registries and published cohort data to translate relative risks and cumulative incidences into an integrated life-expectancy framework, stratified by age at donation, race, and sex, and to separately quantify the life-expectancy value of priority waitlist access, both unconditionally at the population level and conditional on the recipient having actually developed ESRD.**

Methods

Model overview

I constructed a Markov state-transition model to compare lifetime outcomes for two hypothetical cohorts of otherwise identical individuals: one cohort that donates a kidney and one that does not. Both cohorts enter the model at the age of donation in full health, and the simulation advances in annual cycles until age 110. The analysis was conducted from the perspective of the individual donor rather than the health-care system; outcomes are reported as undiscounted remaining life-years, consistent with the convention in burden-of-disease analyses.¹⁸ The base-case age at donation is 40 years; subgroup analyses span ages 25–55. I used a Monte Carlo simulation with 5×10^6 individuals per arm and cross-validated against a deterministic analytic cohort implementation. I used annual (one-year) cycles throughout. The Markov model structure is shown in Figure 1.

Health states

The model contains seven mutually exclusive health states (Figure 1): *Healthy*, *ESRD onset*, *Dialysis*, *Standard waitlist*, *Priority waitlist*, *Post-transplant*, and *Dead* (absorbing). At ESRD onset, the cohort branches: a fraction is listed preemptively (bypassing dialysis entirely) and the remainder initiates hemodialysis. The donor arm routes ESRD survivors to the *Priority waitlist*, reflecting the allocation priority afforded to prior living donors (PLD) under KAS250; the non-donor arm routes to the *Standard waitlist*. Graft failure returns patients to the *Dialysis* state, bypassing the ESRD-onset branch.

ESRD risk after donation

Fifteen-year cumulative ESRD incidence was taken from Muzaale et al.⁴: 30.8 per 10,000 (95% CI 24.3–38.5) for donors and 3.9 per 10,000 (95% CI 0.8–8.9) for controls matched by age, sex, and health (8-fold relative risk). Race-specific donor rates were 74.7/10,000 (Black) and 22.7/10,000 (White).⁴ The White non-donor baseline was taken from Grams et al.¹⁹ (sex-averaged 0.05%/15 yr), because zero ESRD events were recorded in White matched controls in Muzaale et al.

I modeled ESRD hazard with a Weibull distribution. I fit the shape parameter ($k = 2.54$) from the five cumulative-incidence-of-ESRD curves (median, interquartile range, and 1st/99th percentiles) that Massie et al. report at 5, 10, 15, and 20 years post-donation.⁵ The Weibull CDF linearizes under the complementary log-log transform, $\ln(-\ln(1 - I(t))) = k \ln(t) - k \ln(\lambda)$, so ordinary least-squares regression of the transformed incidence on $\ln(t)$ recovers k directly as the slope; applied to the median curve this gives $k = 2.54$, reproducing all four published points to within 5%. Fitting the same regression separately to each of the other four published curves gives k ranging 2.15–2.54, confirming a single accelerating Weibull shape ($k > 1$) is broadly consistent across the reported risk distribution; a constant-hazard ($k = 1$) model is ruled out, as it would predict only $\approx 21/10,000$ at 20 years from the observed 15-year rate of 16/10,000, versus the 34/10,000 actually reported. Once fit, k was combined with a per-subgroup Weibull

scale λ , calibrated via a competing-risk bisection algorithm to reproduce each subgroup's 15-year cumulative incidence while accounting for background-mortality attrition in the at-risk pool. I drew within-donor subgroup hazard ratios from Massie et al.: 2.96 (95% CI 2.25–3.89) for Black race, 1.88 (95% CI 1.50–2.35) for male sex, and 1.40 per decade (non-Black donors).⁵

Dialysis survival and waitlist listing

I stratified dialysis mortality by age. I took adjusted mortality among prevalent adult dialysis patients from the USRDS 2023 Annual Data Report (ADR) and converted rates to annual probabilities under a constant hazard: 6.8%/year (ages 18–44), 11.4% (45–64), 17.8% (65–74), and 25.1% (≥ 75).²⁰ I used 2019 data rather than those from 2020–2021 because COVID-19 raised dialysis mortality by roughly 20% in those years. I multiplied first-year mortality by 1.06, the ratio of year-1 to year-2 mortality in the 2019 incident hemodialysis cohort.²⁰ As a check, these rates reproduce USRDS remaining life expectancy on dialysis within 15% for every 5-year age group from 40–44 to 80–84.

At ESRD onset, a fraction of patients is listed preemptively before dialysis initiation, avoiding dialysis mortality entirely. For non-donors, I drew the preemptive listing probability from USRDS 2025 ADR: 5.8% overall.²¹

I derived the conditional annual probability of non-donors on dialysis joining the waitlist from USRDS 2025 ADR.²¹ Approximately 25% of 18–44 year-old ESRD patients are listed or transplanted within one year, with $\approx 9.4\%$ of the ESRD cohort listed preemptively, so the conditional post-dialysis listing rate is $(25\% - 9.4\%)/(100\% - 9.4\%) \approx 17\%$ /year. Donor-arm values for both parameters are described in the next subsection.

Outcomes after ESRD in prior donors

Registry inputs describe all ESRD patients and transplant candidates, not prior donors. Prior donors who develop ESRD are closely followed, are generally healthier, and are listed earlier than other ESRD patients.¹⁴ I therefore gave the donor arm donor-specific post-ESRD inputs from Muzaale et al., who followed 99 US living donors who developed ESRD (median age at onset 50) and compared them with matched non-donors with ESRD.¹⁴

Two inputs were taken directly from that study. First, 21 of 99 donors (21.2%) avoided dialysis at onset (20 preemptive listings and one preemptive living-donor transplant), replacing the general preemptive-listing probability. Second, post-transplant mortality was multiplied by 0.7, the adjusted hazard ratio for prior donors versus matched non-donor recipients (95% CI 0.2–2.4).

Two further inputs were calibrated to the same study, with the cohort entering ESRD at age 50. For each candidate multiplier on donor dialysis mortality, I solved for the annual listing probability that reproduces the observed 43.4% of donors listed within 12 months of ESRD. I kept the multiplier that minimized squared error against observed all-cause mortality 1, 3, 5, and 10 years after ESRD onset (9.1%, 20.2%, 25.5%, and 32.4%). The fit gave a dialysis-mortality multiplier of 0.78 and a listing probability of 0.31/year. The calibrated donor pathway predicts 7.6%, 17.2%, 23.8%, and

37.2% mortality at 1, 3, 5, and 10 years, respectively, all within the published 95% CIs.

Transplant waiting time and waitlist outcomes

Transplant probability while listed was modeled as a time-homogeneous exponential process with rate $r = 1/\mu$, where μ is the mean waiting time, the natural parameterization of the exponential distribution.²² For the standard waitlist, I derived μ from SRTR 2023 ADR (2018–2020 listing cohort, 3-year outcomes) using the same competing exponential model applied to the removal rate: with total exit rate $\Sigma\lambda = -\ln(P_{\text{still}})/3 = 0.403/\text{year}$ and transplant-specific rate $\lambda_T = (P_{\text{tx}}/P_{\text{exit}}) \times \Sigma\lambda = 0.255/\text{year}$ (where $P_{\text{tx}} = 0.443$ accounts for both DDKT and LDKT), the implied mean is $\mu = 365.25/\lambda_T = 1,437$ days.¹ Using this SRTR-calibrated value makes the transplant probability internally consistent with the competing removal rate, which was derived from the same figure. The one-way sensitivity range tests 24 months (730 days) and the pre-KAS250 historical mean of 1,857 days²³ (Table 1); the Punjala 2024 post-KAS250 mean of 1,765 days²³ provides a separate registry-based reference estimate. For the PLD priority waitlist, I used the post-KAS median of 102.6 days,¹⁵ converted to a mean via $\mu = \text{median}/\ln 2 \approx 148$ days. I stratified waitlist mortality by age using 2019 pretransplant mortality among adult kidney candidates from SRTR 2023 ADR, chosen to match the USRDS dialysis year: 1.98 (ages 18–34), 2.95 (35–49), 4.97 (50–64), and 7.43 (≥ 65) deaths per 100 patient-years.¹ Competing removal (delisting for reasons other than transplant or death) was derived from the 3-year cumulative removal incidence of 19.06% reported in SRTR 2023 ADR Figure KI 22, back-calculated to a cause-specific annual rate of 11.4%/year via competing-risk bisection accounting for simultaneous attrition from transplantation and death.¹

Post-transplant survival

I derived post-transplant annual mortality from age-stratified 5-year Kaplan–Meier estimates for DDKT recipients in the SRTR 2023 ADR (Figure KI 70, 2016–2018 cohort): 0.9%/year (ages 18–34), 1.8%/year (35–49), 3.9%/year (50–64), and 6.9%/year (≥ 65).¹ The base case uses DDKT survival because prior living donors who develop ESRD predominantly receive transplants via the deceased-donor priority pathway rather than living-donor transplantation, consistent with Muzaale et al., who found that prior donors predominantly receive deceased-donor kidneys (87%) rather than living-donor transplants.¹⁴ A one-way sensitivity analysis substitutes LDKT patient survival (SRTR 2023 ADR Figure KI 76: 0.4%, 0.8%, 1.7%, and 3.9%/year by age band) to bound the effect of this assumption. Annual graft failure post-year 1 was likewise age-stratified, derived as $1 - (S_5/S_1)^{1/4}$, where S_n denotes the n -year graft survival probability, from age-specific 1- and 5-year DDKT graft survival in the SRTR 2023 ADR (Figure KI 53, 2016–2018 cohort): 3.9%/year (ages 18–34), 3.4%/year (35–49), 4.9%/year (50–64), and 7.8%/year (≥ 65).¹ Graft failure returned patients to the Dialysis state. In the donor arm, post-transplant mortality was further scaled by the prior-donor hazard ratio.

Age dependence and background floor for post-ESRD states

All age-banded post-ESRD inputs (dialysis, waitlist, and post-transplant mortality, and graft failure) were assigned to the midpoint of each band and interpolated log-linearly between midpoints, held constant beyond the outermost midpoints. The lower-boundary bands (18–34 and 18–44) and the upper open-ended bands (≥ 65 and ≥ 75) were assigned representative ages of 26, 31, 70, and 80 to avoid artifacts caused by step functions at band edges.

In every post-ESRD state, annual mortality was also floored at the general-population life-table probability for the same age. Without this floor, state-specific rates fall below background mortality at advanced ages (e.g., 6.9%/year post-transplant mortality at ages ≥ 65 is exceeded by background mortality from age 83), which would make ESRD paradoxically protective late in life.

Background mortality and donor all-cause survival

I took background mortality in the Healthy state from the 2021 US Life Tables,²⁴ with sex-specific tables used in sex-stratified analyses. As discussed in the Introduction, the published literature on donor all-cause mortality does not consist of noisy estimates of one fixed hazard ratio (HR): short-to-medium-follow-up cohorts and the meta-analysis pooling them report no excess mortality,⁶ while the one cohort followed long enough to plausibly detect a late-emerging effect reports a significantly elevated HR,⁹ with survival curves separating only after approximately ten years.^{10,11} I therefore modeled the donor all-cause mortality HR as time-varying rather than fixed: flat at 1.0 (no excess risk) for the first 10 years after donation, then increasing linearly to 1.30 by 15 years post-donation, matching the Mj  en et al.⁹ cohort’s median follow-up, and held flat at 1.30 thereafter. The non-donor arm’s background mortality HR is fixed at 1.0 throughout. Because this reflects a genuine question about model structure (does excess mortality risk exist at all, and if so, when does it emerge) rather than sampling variation around an agreed-upon value, it is not included in the probabilistic sensitivity analysis; instead, I report a dedicated structural sensitivity analysis over the plausible alternative HR assumptions in the literature.

Subgroup analyses

I repeated the analyses at ages 25, 30, 35, 40, 45, 50, and 55 years at donation. Donor ESRD rates were scaled by the Massie 2017 age HR (1.40/decade, non-Black donors); Massie et al. found no significant age gradient among Black donors (HR 0.88/decade, 95% CI 0.72–1.09), so Black donor rates were held constant across age.⁵ Non-donor baselines were scaled by the age gradient of Grams et al.’s 15-year projections for donor-like candidates at ages 20, 40, and 60, interpolated log-linearly by race and sex.¹⁹ These projections show Black non-donor risk rising steeply from age 20 to 40 but only ≈ 1.13 -fold per decade from 40 to 60, against ≈ 1.45 -fold for White candidates. The “overall” non-donor arm used in the age sweep keeps its Muzaale matched-control anchor at age 40 and uses the White gradient, since White donors make up most of the donor pool.

I took race-specific donor ESRD rates (Black 74.7/10,000; White 22.7/10,000) and race-specific non-donor baselines (Black sex-averaged 0.195%/15 yr; White sex-averaged 0.05%/15 yr, both at age 40) from Muzaale et al.⁴ and Grams et al.,¹⁹ respectively. Race-specific post-transplant survival was from SRTR 2023 Figure KI 71. For waitlist mortality, the age-stratified rates were multiplied by the ratio of race-specific to overall 2019 pretransplant mortality (Figure KI 26; Black 0.94, White 1.17), assuming race and age act multiplicatively.¹

I derived sex-specific donor ESRD rates by applying the Massie 2017 male-sex HR (1.88)⁵ to the overall donor rate, weighted by the observed donor sex mix (63.5% female; SRTR 2023 ADR Figure KI 104).¹ For non-donor baselines, I used the Grams 2016 White sex-specific rates (female 0.04%/15 yr; male 0.06%/15 yr at age 40, scaled by the sex-specific Grams age gradient)¹⁹ rather than applying the within-donor sex HR to the control arm, because Massie 2017 hazard ratios describe variation within the donor cohort and are not appropriate for scaling a different reference population. I applied this sourcing rule, Muzaale matched-control rates for overall analyses and Grams direct rates wherever sex or race is stratified, with the Grams age gradient applied to every non-donor baseline, consistently across all simulations.

Probabilistic sensitivity analysis (PSA)

I conducted a PSA with 2,000 parameter draws. Sampled quantities and their distributions were:

- ESRD 15-year cumulative risks: beta distributions parameterized from the Muzaale 2014 published 95% CIs;⁴
- Weibull shape k : log-normal, $\sigma = 0.069$, the sample standard deviation of $\ln k$ across the five independent cloglog-regression fits described above (Massie et al. does not report a CI for this parameter directly);
- Waitlist listing probability: beta, fractional SE = 40%, consistent with the 0.05–0.30 plausible range; and
- Prior-donor post-ESRD inputs from Muzaale et al. (2016):¹⁴ preemptive fraction, beta with binomial SE (19% of the mean); listing probability, beta with fractional SE = 23%, from the 95% CI of the proportion listed by 12 months; dialysis-mortality multiplier, log-normal with $\sigma = 0.234$, from the 95% CI of the overall mortality HR after ESRD (0.4–1.0); and post-transplant multiplier, log-normal with $\sigma = 0.634$, from its 95% CI (0.2–2.4).

Registry-derived parameters (from SRTR or USRDS) have negligible sampling error relative to structural uncertainty and were fixed at base-case values; scenario variation in these parameters is captured by the one-way sensitivity analysis. The donor all-cause mortality HR is deliberately excluded from the PSA: as described above, the disagreement among published estimates reflects a structural question about the time-course of donor risk, not sampling error around a single value, and is instead addressed by a separate, dedicated structural sensitivity analysis. I report the mean Δ LE, 95% credible interval, and the probability that donation is beneficial to life-expectancy.

One-way sensitivity analysis

I varied each parameter individually, holding the donor all-cause mortality HR at its base-case time-varying profile:

- No priority access and changing the PLD wait time
- ESRD relative risk from the Mj  en et al. point estimate on ESRD HR (11 $\times$ controls)
- Post-transplant survival at LDKT quality
- Dialysis mortality $\pm 50\%$
- Prior-donor post-ESRD inputs replaced with the general ESRD population's (preemptive listing, listing rate, and dialysis and post-transplant mortality)
- Donor dialysis-mortality multiplier at the bounds of the Muzaale et al. mortality HR (0.4 and 1.0)

Structural sensitivity to donor mortality hazard ratio

Because the donor all-cause mortality HR is not sampled in the PSA or perturbed in the one-way sensitivity analysis above, I separately compared the base-case time-varying HR profile against the flat (time-invariant) HR implied by individual published estimates, taken at face value for the donor's entire remaining lifetime: HR = 1.0 (no excess mortality), HR = 1.30 (the Mj  en et al. point estimate),⁹ and HR = 1.52 (the upper bound of the Mj  en et al. 95% CI). This analysis is intended to bound how much the result depends on which single-HR assumption is adopted, rather than to imply that any of these flat alternatives is itself a plausible model of donor risk.

Voucher benefit

A kidney-donation "voucher" is a mechanism by which a living donor designates up to five healthy, non-donor family members to receive PLD priority-waitlist access should any one of those family members ever develop ESRD (a voucher used by one family member expires the vouchers of the other family members). I simulated two arms of healthy non-donors, both carrying non-donor baseline ESRD risk and a background all-cause mortality HR of 1.0. The arms differed in both waitlist priority and post-transplant survival type. The voucher arm used a priority mean wait of 148 days (the post-KAS250 median of 102.6 days from Wainright et al.¹⁵ converted via $\mu = \text{median} / \ln 2$; this estimate comes from a study of KAS250 deceased-donor access for prior living donors and is used here as a proxy for the shorter-than-standard wait enabled by an NKR living-donor chain), together with LDKT post-transplant patient survival (SRTR 2023 ADR Figure KI 76: 0.4%, 0.8%, 1.7%, and 3.9%/year at ages 18–34, 35–49, 50–64, and ≥ 65) and LDKT graft survival (Figure KI 61), because NKR vouchers are designed to facilitate living-donor kidney exchange chains. The control arm used the SRTR-calibrated standard mean wait of 1,437 days¹ and DDKT post-transplant survival, identical to the base-case non-donor arm. I assigned the voucher arm the USRDS preemptive-listing probability for ages 18–44 (9.4%), as a proxy for an informed, monitored family member, and the control arm the general-population probability (5.8%).²¹ All other transition probabilities (dialysis mortality, per-cycle waitlist listing, waitlist mortality and removal) were identical

to the base-case non-donor arm; because voucher holders are not donors, neither arm used the prior-donor inputs. I ran the analysis at the base-case age of 40 with an age sweep and an ESRD-risk subgroup analysis using the Muzaale overall and Grams race-specific non-donor cumulative incidences, scaled to each entry age by the Grams age gradient.^{4,19}

Because the voucher (and, by extension, the donor PLD-priority) benefit is diluted at the population level by the large fraction of the cohort who never develop ESRD, I ran a second model conditioned entirely on ESRD onset. All individuals entered the model already diagnosed with ESRD in order to isolate the life-expectancy benefit of priority waitlist access for the sub-population who actually reach the waitlist. Entry to dialysis versus preemptive listing (9.4% priority arm, 5.8% standard arm, as in the voucher analysis), dialysis and waitlist mortality, waitlist removal, and background mortality used the same age-dependent parameters as the base-case non-donor arm, including the life-table floor; I included no separate Healthy state. I compared two arms differing in both waitlist priority and post-transplant survival: the priority arm used the priority mean wait (148 days) and LDKT patient and graft survival (the same inputs as the voucher arm above), while the standard arm used the standard mean wait (1,437 days) and DDKT survival. The base case used an ESRD-onset age of 60 with an age sweep.

Results

Base-case life expectancy

In the base-case analysis, kidney donation was associated with a life-expectancy cost of $\Delta LE = -891.3$ days. Across all base-case and subgroup scenarios, donation was associated with a net life-expectancy cost to the donor under this base-case HR assumption.

Racial disparity

Race was a comparatively minor modifier of life-expectancy cost once the donor all-cause mortality HR, which affects all donors equally regardless of race, dominates the base-case result (Figure 2). At age 40, Black donors experienced a cost of 896.5 days, about 2% larger than the 877.3 days estimated for White donors. This gap was wider at younger donation ages and closed at older ages: a Black donor at age 25 loses 1,246.9 days of life expectancy compared with 1,069.7 days for a White donor of the same age (a 16% difference), while at age 55 the two estimates are essentially equal (583.5 vs. 587.0 days). This disparity at younger ages traces to post-donation ESRD incidence: Black donors have a 15-year cumulative incidence of 74.7 per 10,000 versus 22.7 per 10,000 in White donors,⁴ a 3.3 $\times$ difference further compounded by a higher background ESRD rate in age- and health-matched Black non-donors.¹⁹ The closing of the gap at older donation ages reflects a significant age $\times$ race interaction reported by Massie et al.: among non-Black donors, post-donation ESRD risk rises with age at donation (HR = 1.40/decade, 95% CI 1.23–1.59), but among Black donors, no age gradient was detected (HR = 0.88/decade, 95% CI 0.72–1.09).⁵ Non-donor projections show a weaker version of the same pattern, with Black risk flattening after

age 40.¹⁹ As a result, Black donors' excess 15-year ESRD incidence over matched non-donors narrows from 66 to 51 per 10,000 between ages 25 and 55, while White donors' widens from 12 to 29 per 10,000. Massie et al. note this flat age effect in Black donors may partly reflect a selection effect at donor evaluation, wherein older Black candidates predisposed to CKD are more likely to be screened out before donation, rather than an absence of age-related risk in the general population.⁵

Age at donation

The life-expectancy cost decreased monotonically with older age at donation, from 1,092.6 days at age 25 to 592.4 days at age 55, roughly 84% larger at the youngest age relative to the oldest (Figure 2). Unlike the racial and sex disparities, this age gradient is driven primarily by the time-varying mortality HR rather than by ESRD risk: a younger donor lives long enough after the ten-year onset window for a much larger share of their remaining lifetime to fall under the elevated (up to 1.30) hazard ratio, whereas an older donor's remaining life expectancy is increasingly concentrated in the years before the HR has fully risen.

Sex

Male and female donors faced broadly similar life-expectancy costs at age 40 (891.6 days for male donors vs. 872.5 days for female donors, a difference of about 2%), even though male donors carry a within-donor ESRD hazard ratio of 1.88 relative to female donors.⁵ The sex-specific ESRD risk difference is a minor contributor next to the much larger cost imposed by the time-varying mortality HR, which is applied identically to male and female donors: a Black male donor at age 40 loses 902.2 days, about 1% more than the 891.6 days for an overall male donor and about 5% more than the 862.2 days for a White female donor of the same age (Figure 2). At age 25, Black male donors lose 1,291.3 days versus 1,026.4 days for White female donors, a 26% difference that is larger than at age 40 for the same reason the racial disparity is larger at younger ages.

Probabilistic sensitivity analysis

The PSA of 2,000 parameter draws (Figure 3), which holds the donor all-cause mortality HR fixed at its base-case time-varying profile, produced a mean ΔLE of -896.7 days (95% credible interval -941.4 to -863.0 days; $P[\text{donation beneficial}] = 0\%$). With the HR held fixed, the residual parameter uncertainty captured by the PSA (ESRD incidence, Weibull shape, waitlist listing probability, and prior-donor post-ESRD inputs) is narrow and consistently negative.

One-way sensitivity analysis

With the donor all-cause mortality HR held fixed at its base-case time-varying profile rather than perturbed, no single parameter in the one-way sensitivity analysis produced a swing exceeding 35 days (Table 1, Figure 4). The largest was replacing the prior-donor post-ESRD inputs with those of the general ESRD population (-32.7 days), followed by the donor ESRD relative risk ($+33.0$ days at $2\times$; $+21.7$ days

at $4\times$), the donor dialysis-mortality multiplier (+24.5 days at 0.4), and dialysis mortality $\pm 50\%$ (−16.1 to +20.1 days). All remaining parameters, including PLD and standard waitlist wait times and post-transplant graft quality, shifted Δ LE by under 10 days.

Structural sensitivity to donor mortality hazard ratio

I separately compared the base-case time-varying HR profile against the flat HR implied by individual published estimates, applied for the donor's entire remaining lifetime (Figure 5). The textbook “no excess mortality” assumption (HR = 1.0 flat) yields a cost of only 55.0 days, an order of magnitude smaller than the time-varying base case. The base-case time-varying profile (1.0 ramping to 1.30 over years 10–15) yields −891.3 days. Applying the Mj  en et al. point estimate (HR = 1.30) from the moment of donation, rather than only after the ten-year lag their own data support, yields −1,066.5 days. The upper bound of the Mj  en et al. 95% CI (HR = 1.52), applied flat, yields −1,674.2 days. This 1,619-day range, from a cost of under two months to a cost exceeding four years, dwarfs every other source of uncertainty examined in this analysis and is the dominant open question for any model of donor life expectancy.

Donor priority benefit

Although KAS250 affords prior living donors (PLD) priority access to the deceased-donor waitlist,¹⁵ the resultant life-expectancy benefit to the donor is small at the population level. Removing PLD priority access entirely, setting the donor waitlist to the standard 1,437-day mean wait, worsened Δ LE by 6.2 days relative to base case. Reducing the priority wait to 50 days (median, equivalent to a mean of ≈ 72 days) had negligible additional effect. This insensitivity to the exact priority wait time arises because approximately 0.31% of donors develop ESRD within 15 years and fewer still ever reach the transplant waitlist; the priority pathway is simply too rarely invoked to move a population-level outcome meaningfully.

Voucher benefit

The life-expectancy benefit of a kidney-donation voucher to a healthy, non-donor family member was small: +1.92 days at age 25, +1.47 days at age 40, and +0.34 days at age 60. This reflects the same dilution mechanism: conditional on developing ESRD and reaching the priority waitlist, priority access via an NKR living-donor chain adds a clinically substantial +1,488 days of life expectancy for a patient whose ESRD onset occurs at age 40, and +608 days for onset at age 60, declining steadily with onset age (Figure 6). However, since only approximately 0.04% of non-donors develop ESRD within 15 years, this large conditional benefit compresses to under two days at the population level. Voucher benefit was higher for Black non-donor family members (+7.30 days at age 40), owing to their approximately four-fold higher baseline ESRD incidence.¹⁹

Discussion

Under the base-case time-varying hazard assumption, the mean life-expectancy cost to the individual donor (≈ 891 days, or 2.4 years) must be weighed against the benefit conferred on the recipient and, through the voucher mechanism, on designated family members who may one day develop ESRD. Contemporary studies estimate kidney transplant candidates experience a gain of five life years from transplantation.²⁵ Viewed in this frame, donation represents a transfer of life expectancy from donor to recipient on favorable terms.

This aggregate figure conceals some heterogeneity. Age at donation is the demographic factor that matters most: the life-expectancy cost is roughly 84% larger at age 25 relative to age 55, because a younger donor lives long enough after the roughly decade-long delay in donor mortality risk for a much larger share of their remaining life to fall under the elevated hazard ratio. Race and sex differ by about 2% each at the base-case donation age; the racial difference grows to 16% at age 25 and disappears by age 55. This analysis supports age-, race-, and sex-specific life-expectancy disclosure: a 25-year-old Black male and a 55-year-old White female donor are, in effect, being asked to accept quantitatively different bargains under the same generic counseling framework.

The model also reframes what priority-access policy is actually protecting. KAS250 and the NKR voucher mechanism are designed around a rare event, an eventual ESRD diagnosis in a prior donor or designated family member, and this analysis shows that their unconditional, population-level life-expectancy value is correspondingly small because so few beneficiaries ever draw on it. This is not evidence that such policies are unimportant: conditional on the rare event they are designed for, priority access can be worth years at the individual level, and the dilution reflects low incidence and effective screening rather than low value per event. The appropriate way to communicate these mechanisms may therefore be closer to catastrophic insurance, whose expected value is small precisely because it is rarely invoked, than to an intervention expected to move average outcomes.

The results are dominated by a single structural uncertainty, the time-course of the donor all-cause mortality hazard ratio (HR), rather than by sampling error in any individually estimated parameter. Short-to-medium-follow-up cohorts and the meta-analysis pooling them report no excess mortality, while the one cohort followed long enough to plausibly detect a late-emerging effect reports a hazard ratio of 1.30 that its own authors attribute to risk emerging only after roughly a decade. This analysis encodes that pattern as a piecewise-linear ramp (flat at 1.0 through year 10, rising to 1.30 by year 15, matching the Mj  en et al. cohort's median follow-up), which is a minimal, directly literature-motivated functional form but an assumption nonetheless. The true magnitude and time-course could differ. If the true HR is closer to 1.0, race, sex, and age play far greater roles.

A natural question is why Mj  en's finding has not already revised donor counseling practice. O'Keeffe et al. pooled four available cohort studies of long-term donor mortality

relative to healthy matched controls. Of the four, only Mj  en followed respondents for more than 10 years, and only Mj  en reported an HR greater than 1.0. O’Keeffe’s pooled conclusion was “no evidence that living kidney donors had higher risk for mortality.”⁶ But the studies are not contradictory: Mj  en argued,¹⁰ and O’Keeffe agreed,¹¹ that the survival curves in Mj  en et al. only began to separate after approximately ten years of follow-up, so the three shorter-follow-up cohorts were not long enough to detect the effect. Current practice reflects the O’Keeffe pooled conclusion, which is the best available evidence synthesis but is dominated by studies whose follow-up windows end before the separation Mj  en identified. This paper does not adjudicate between the competing HR estimates; it quantifies what is at stake in that adjudication, making the practical consequences of the literature’s unresolved question legible to clinicians and to donors themselves, regardless of which estimate ultimately proves correct.

I therefore join Mj  en’s call¹⁰ for Garg,⁷ Segev,³ and their colleagues to “publish updated results from their study cohorts,” which could now span 22 years of follow-up, becoming the longest large-scale studies of donor all-cause mortality ever conducted. Future work should prioritize prospective donor cohorts with follow-up long enough, ideally beyond 15–20 years, to directly estimate whether and when donor all-cause mortality risk begins to rise, since **this single question now determines the dominant magnitude of the life-expectancy cost.**

Acknowledgments, Statements, and Declarations

Ethical Considerations

This study analyzed only publicly available, de-identified, aggregate data from published cohort studies and national registry reports (USRDS, SRTR/OPTN Annual Data Reports); no primary human-subjects data were collected. Institutional Review Board approval was therefore not applicable.

Consent to Participate

Not applicable. This study did not involve primary data collection from human participants.

Consent for Publication

Not applicable.

Declaration of Conflicting Interests

The author is a living kidney donor; that experience motivated this study. The author declares no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Data Availability

All parameter sources are published, publicly available studies and registry reports, cited throughout the manuscript. The simulation code and parameter-assembly pipeline are implemented in Python 3.10 (NumPy, pandas, matplotlib) and openly available in a public repository at <https://github.com/rupumped/kidney-donor-le>, consistent with Level 1 of the Transparency and Openness Promotion (TOP) Guidelines. No restricted-access data were used.

Demographic Classification Variables

Race is used in this analysis as reported in the underlying data sources rather than as a directly measured biological quantity. Donor-side race data^{1,4,5,21,26} derive from categories reported to the Organ Procurement and Transplantation Network (OPTN) at donor evaluation and registration; the non-donor comparator population used by Muzaale et al.⁴ was drawn from the Third National Health and Nutrition Examination Survey (NHANES III), in which race is self-reported by participants. Race is included in the model because it is the variable available in these source populations and because it is strongly associated with measured post-donation ESRD incidence. Part of this association has an identified biological correlate, *APOL1* renal-risk-variant genotypes, which are more common among individuals of recent African ancestry and are directly implicated in donor kidney-disease risk.¹² However, socioeconomic status, healthcare access, comorbidity burden, and other factors correlated with self-reported race in the United States were not available in the source registries and are not separately modeled here. The race-stratified life-expectancy estimates reported in this paper should therefore be interpreted as reflecting the net effect of these unmeasured, race-correlated factors together with any effect of race itself, rather than race operating as an isolated causal variable.

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Tables

Table 1. One-way sensitivity analysis: parameter values tested and the resulting change in life expectancy (donor arm minus non-donor arm), age 40 at donation, all other parameters held at base-case values, including the donor all-cause mortality HR, which is held at its base-case time-varying profile rather than perturbed here (see Table 2 and Figure 5 for sensitivity to that assumption). Negative Δ LE values indicate a net life-expectancy cost of donation relative to the matched non-donor arm under the value tested; positive values indicate a donor-arm benefit. The “8× controls (base case)” row reproduces the overall base-case result reported in the main results (Δ LE = −891.3 d, analytic cohort).

Parameter (value tested)	Δ LE (days)	vs. base (days)
<i>ESRD relative risk, donor vs. non-donor (base case: 8×, 30.8/10,000)</i>		
2× controls	−858.3	+33.0
4× controls	−869.6	+21.7
8× controls (base case)	−891.3	0.0
11× controls (Mjølén et al. point estimate)	−908.4	−17.1
<i>Standard waitlist mean wait (base case: 1,437 d)</i>		
24 months (730 d)	−891.5	−0.2
61 months (1,857 d, pre-KAS250) ²³	−891.3	0.0
<i>Priority (PLD) waitlist mean wait (base case: $\approx$ 148 d, from 102.6 d median via $\mu = \text{median}/\ln 2$)</i>		
50 d median ($\approx$ 72 d mean, optimistic)	−891.1	+0.2
200 d median ($\approx$ 289 d mean)	−892.2	−0.9
No priority access (PLD = standard, 1,437 d mean)	−897.5	−6.2
<i>Dialysis mortality (base case: USRDS 2019 age-specific rates)</i>		
−50%	−871.2	+20.1
+50%	−907.4	−16.1
<i>Post-transplant survival (base case: DDKT quality)</i>		
LDKT quality (SRTR Figure KI 76)	−883.7	+7.6
<i>Prior-donor post-ESRD inputs (base case: Muzaale et al. 2016)</i>		
General ESRD population inputs	−924.0	−32.7
Donor dialysis multiplier 0.4 (base 0.78)	−866.8	+24.5
Donor dialysis multiplier 1.0	−902.8	−11.5

Table 2. Structural sensitivity: donor mortality hazard ratio. Δ LE under the base-case time-varying HR profile versus a flat HR taken from each individual published estimate and applied for the donor’s entire remaining lifetime, age 40 at donation. Unlike Table 1, this range reflects a genuine question about model structure (whether, and when, donor mortality risk rises) rather than sampling uncertainty.

HR assumption	Δ LE (days)
HR = 1.00 flat (no excess mortality)	−55.0
HR 1.00 $\rightarrow$ 1.30, years 10–15 (base case)	−891.3
HR = 1.30 flat (Mjølén et al. 2014 point estimate)	−1,066.5
HR = 1.52 flat (Mjølén et al. 2014 upper 95% CI)	−1,674.2

Figures

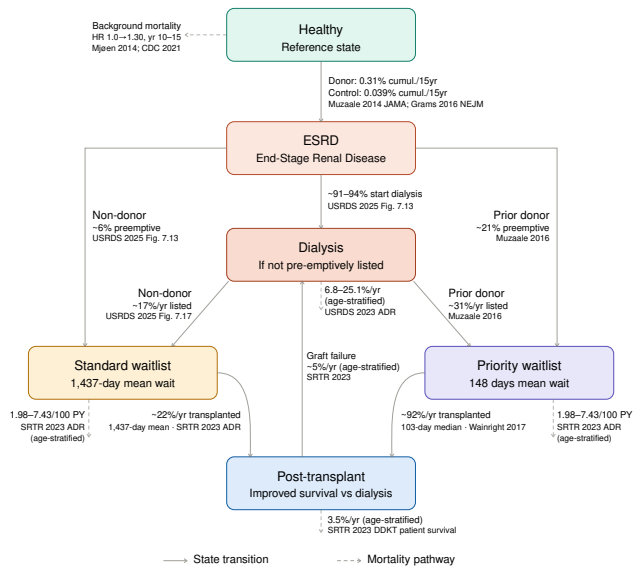


Figure 1. Markov state-transition model for kidney donation. Boxes represent the seven mutually exclusive health states; arrows show permitted transitions. The donor arm (upper pathway) routes ESRD-onset patients to the Priority waitlist; the non-donor arm routes to the Standard waitlist. The Dead state is absorbing.

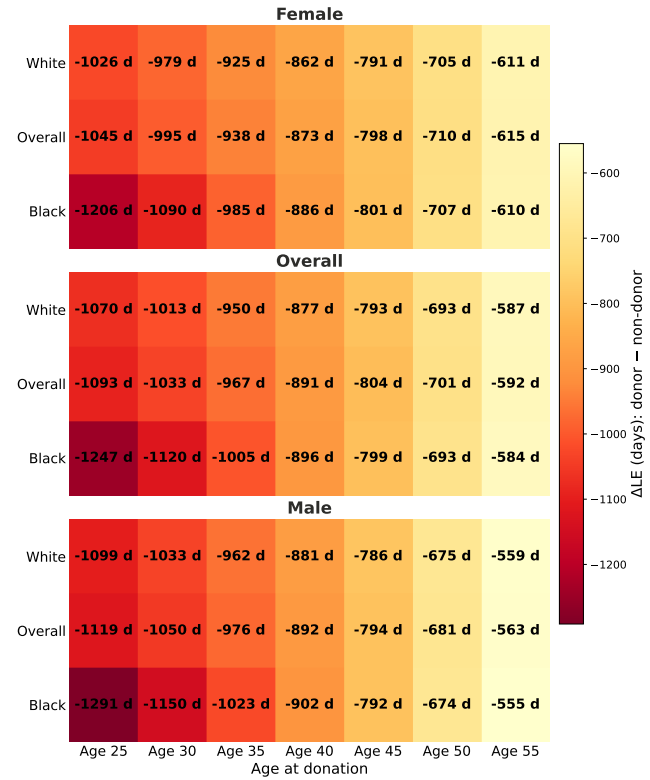


Figure 2. Life-expectancy impact of kidney donation by age, race, and sex. Each cell shows ΔLE (donor minus non-donor, days) for the indicated sex panel, race row, and age-at-donation column. **Age at donation dominates:** costs range from 1,291 days for a Black male donor at age 25 to 555 days for a Black male donor at age 55. At age 25, the largest within-age contrast (Black male vs. White female) is 1,291 vs. 1,026 days, respectively, a 26% difference; by age 55, race and sex differences are small.

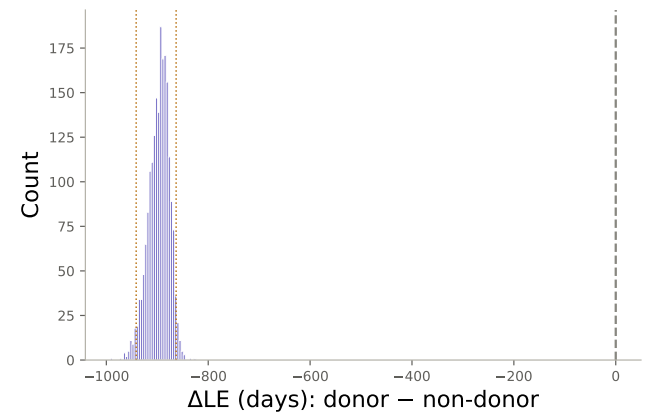


Figure 3. Probabilistic sensitivity analysis (PSA). Histogram of ΔLE (donor minus non-donor, days). The dashed vertical line marks zero; orange dotted lines mark the 95% credible interval bounds.

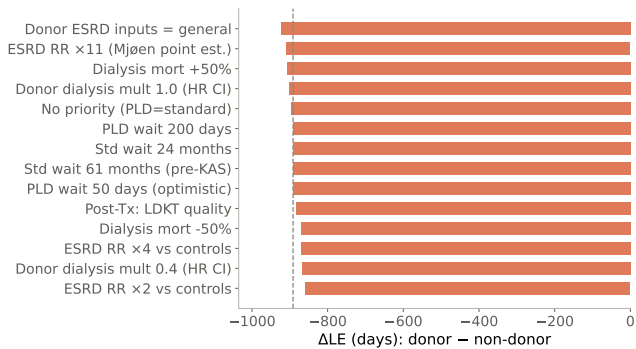


Figure 4. One-way sensitivity analysis: tornado chart. Each bar shows the Δ LE resulting from testing the indicated parameter value, holding the donor all-cause mortality HR fixed at its base-case time-varying profile (see Table 1 for exact values; Figure 5 for sensitivity to the HR assumption itself). Vertical dashed line shows base case. **With the HR held fixed, no parameter shifts Δ LE by more than 35 days; the largest shift comes from replacing prior-donor post-ESRD inputs with those of the general ESRD population.** Priority waitlist parameters (“No priority” and “PLD wait” rows) are barely distinguishable from base case, confirming the negligible population-level effect of transplant priority.

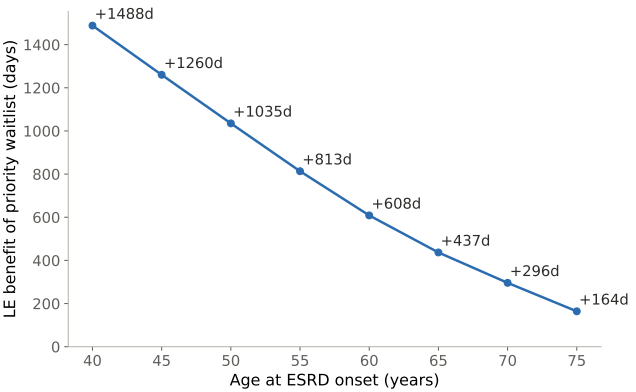


Figure 6. Priority waitlist LE benefit conditional on ESRD onset, by age. Δ LE (priority minus standard waitlist, days) for individuals who have already developed ESRD, as a function of age at ESRD onset. **Conditional on ESRD, priority access is highly valuable: +1,488 days at age 40 onset and +608 days at age 60, declining steadily with onset age.** This large conditional benefit is diluted to under two days at the population level for overall non-donors, and up to +7.3 days for Black non-donor family members, by the low ESRD incidence in non-donors ($\approx 0.04\%/15$ yr) and in donors ($\approx 0.31\%/15$ yr).

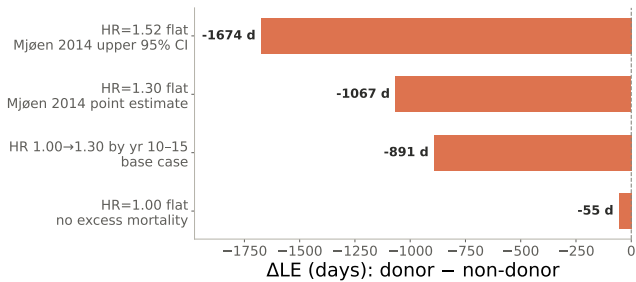


Figure 5. Structural sensitivity to the donor all-cause mortality HR. Each bar shows the Δ LE resulting from replacing the base-case time-varying HR profile (1.0 flat through year 10, ramping to 1.30 by year 15) with a single flat HR held for the donor’s entire remaining lifetime, taken from individual published estimates. **The choice of HR assumption spans a 1,619-day range, from a cost of under two months to a cost exceeding four years, dwarfing every other source of uncertainty examined in this analysis.**